Thermodynamic description of the Mg–Si system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Mg–Si binary system as calculated with pycalphad from the SGTE dataset MgSi-12Sch.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → Mg2Si + (Si), lies at 939.3 °C.

1. Introduction

The Mg–Si system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as congruent compound, simple eutectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of M. Schick, B. Hallstedt, A. Glensk, B. Grabowski, T. Hickel, M. Hampl, J. Groebner, J. Neugebauer, R. Schmid-Fetzer, Calphad, 37, 77-86(2012). The calculations use pycalphad 0.11.2 at 1 atm.

Pure Mg melts at 652.1 °C. For Si, the liquidus meets the axis at 1413.8 °C.

2. Phase diagram

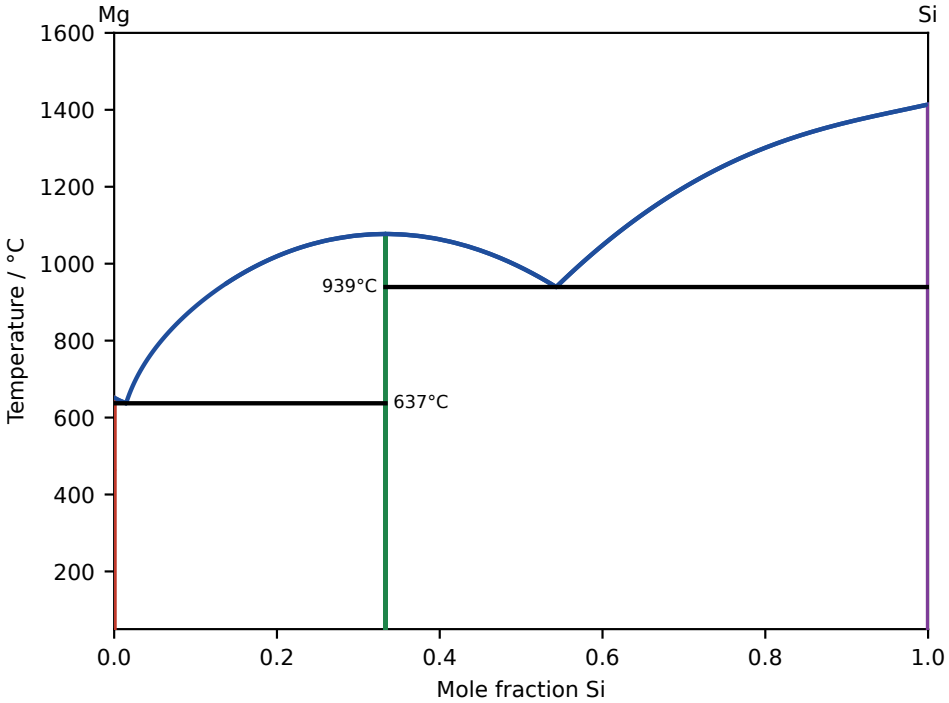


Fig. 1. Calculated Mg–Si phase diagram.

The eutectic reaction L → Mg2Si + (Si) is calculated at 939.3 °C, with L at 54.3 at.% Si. At 637.1 °C the eutectic L → (Mg) + Mg2Si takes place; the L phase then contains 1.5 at.% Si. Mg2Si melts congruently at 1077.3 °C and 33.3 at.% Si.

Table 1. Invariant reactions in the Mg–Si system (calculated).

Reaction	Type	T (°C)	x(Si), 1	x(Si), 2	x(Si), 3
L → Mg2Si + (Si)	eutectic	939.3	0.543	0.333	1.000
L → (Mg) + Mg2Si	eutectic	637.1	0.015	0.000	0.333

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1750 K the integral mixing enthalpy of the liquid has its minimum, −16.34 kJ/mol, at 48.0 at.% Si. The calculated enthalpy of formation of Mg2Si at 298.15 K is −21.05 kJ/mol per mole of atoms.

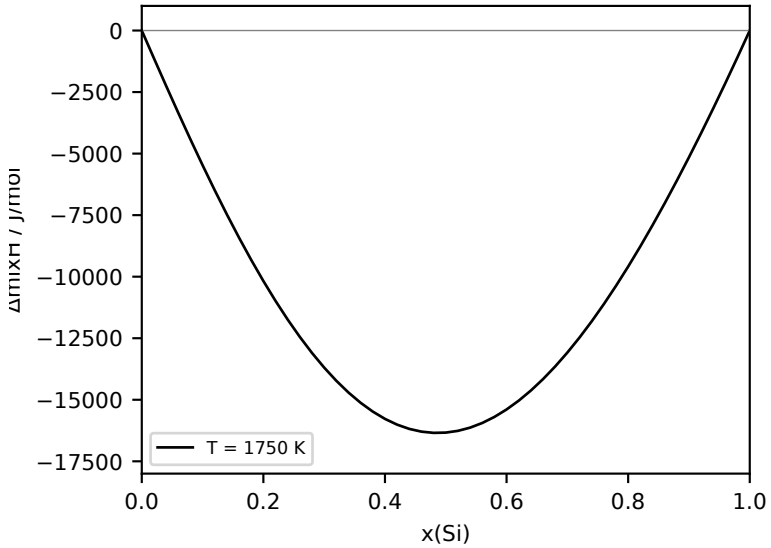


Fig. 2. Calculated mixing enthalpy of the liquid at 1750 K.

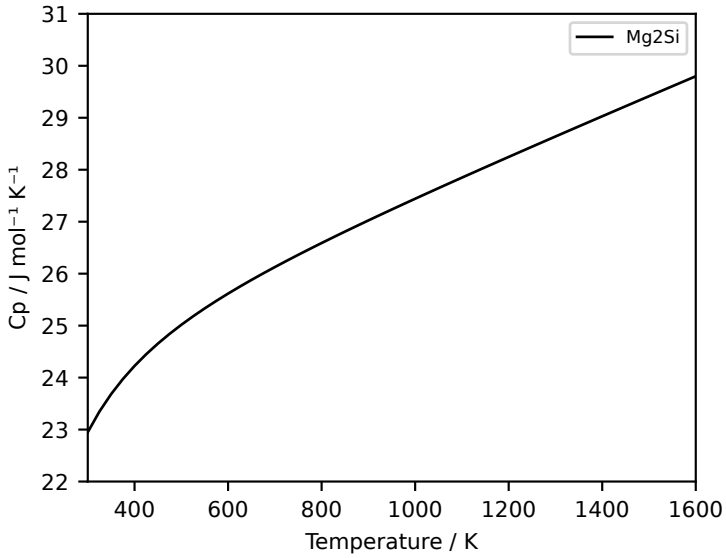


Fig. 3. Calculated heat capacity of Mg2Si.

Table 2. Calculated formation properties of the intermediate phases (Mg–Si).

Phase	at.% Si	ΔfH(298 K) / kJ mol⁻¹	S(298 K) / J mol⁻¹ K⁻¹
Mg2Si	33.3	−21.05 ± 0.5	25.34

Reference: stable elements (SER) at 298.15 K. ± values are synthetic illustration uncertainties, not data.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.